

Buyer Agent Audit & Verification Report

Branch: feature/buyer-agent-verification | **Target:** main | **Repository:** Ritikmehta080905/FarmGenAI | **Status:** Certified (83/83 Tests Passed)

1. Executive Summary

This audit certifies the end-to-end hardening, security guardrails, and user experience enhancements for the **Buyer Agent** and autonomous procurement engine in FarmGenAI. All changes were implemented strictly within the Buyer domain on branch feature/buyer-agent-verification, preserving isolation from non-buyer agents and shared schemas.

2. Audited Findings & Technical Resolutions

Issue 1: Procurement UX Flow Required Redundant Manual Clicks

- Previous:* Submitting requirement modal routed to an idle screen requiring a separate click on "Start Autonomous Top-5 Negotiation".
- Resolution:* Action button updated to **"Submit to AI Validation & Start Negotiation"** with autoStart: true flag. The negotiation room automatically starts runParallelAutonomousNegotiation() upon entry with an active animated streaming status indicator.

Issue 2: Adversarial Input Vulnerability & LLM P_max Overrides

- Previous:* Missing type/boundary assertions exposed the buyer to corrupted inputs (NaN, Inf, negatives) and potential LLM hallucinations accepting offers above reservation ceiling.
- Resolution:* Added strict numeric validation in agents/buyer_agent.py. Deterministic override forcefully rejects or clamps any LLM counter/accept above P_max.

Issue 3: Concurrent Multi-Branch Budget Allocation Leakage

- Previous:* Parallel top-5 seller branches lacked mutual exclusion, risking budget overcommit.
- Resolution:* Implemented thread-safe BudgetReservationTracker with asyncio.Lock() ensuring Committed + Pending + New ≤ Total Budget.

Issue 4: Quantity Lifecycle & Minimum Lot Validation

- Previous:* Candidates with inventory below buyer lot size wasted negotiation rounds.
- Resolution:* Early disqualification when candidate.quantity < min_batch_size; tracks requested, allocated, and remaining quantities.

Issue 5: Listing Freshness & Idempotency at Deal Award

- Previous:* Race conditions where candidate listing was sold/depleted during negotiation rounds.
- Resolution:* Pre-award live DB freshness revalidation and SHA-256 deterministic idempotency_key.

3. Test Suite Verification & Certification (100% Pass)

Test Suite	Coverage Description	Pass / Total	Status
test_buyer_adversarial_pmax_budget.py	NaN/Inf/Negatives/Strings, LLM override, budget lock, quantity lifecycle, freshness	14 / 14	PASS (100%)
test_buyer_runtime_e2e.py	Full runtime E2E chain, no-deal boundary, crop guardrail, invalid quantity	4 / 4	PASS (100%)
test_05_buyer_agent_extensive.py	Buyer agent concessions, personas, scoring, memory, contracts	38 / 38	PASS (100%)
test_07_buyer_guardrail_parallel.py	APMC statutory price caps, parallel auto-selection, floor rejection	5 / 5	PASS (100%)
test_09a_buyer_negotiation_orchestration.py	Top-5 selection, state isolation, landed cost calculation, transcripts	22 / 22	PASS (100%)
TOTAL	Comprehensive Full Regression & Safety Suite	83 / 83	CERTIFIED

4. Pull Request & Deployment Status

- **Active PR:** PR #5 (feat(buyer): direct auto-start negotiation from procurement and enforce P0 budget & Pmax safety)
- **PR URL:** <https://github.com/Ritikmehta080905/FarmGenAI/pull/5>
- **Superseded PR:** PR #4 (Successfully closed)
- **Live Dashboard URL:** <http://localhost:8080/dashboard/buyer>